

Dian Jin

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EDUCATION

John Hopkins University

M.S. in Computer Science.

University of Wisconsin - Madison

B.S. in Computer Science; B.S. in Data Science, 2020 Dean's List.

Baltimore, MD

Aug 2025 - Present

Madison, WI

Aug 2021 - May 2024

PROGRAMMING SKILLS

Languages: Python, C, C++, JavaScript/CSS, C#, Java, HTML, Vue, SQL, Haskell

Field: Embedded System, Operating System, Big Data, Data Structure, Algorithm, Machine Learning, Neural Network, API

Others: Game-play Development, Project Proposals

WORK EXPERIENCE

John Deere

Software Product Engineer

May 2024 - Present

Rotation 2

- **Led full-stack development of a Django web application** that captures real-time data from embedded C++ controllers, transmits it to AWS SQL, and visualizes it via Kinesis streams—reducing latency by 40% and enabling timestamp-based cost analysis.
- **Designed and implemented a cost-efficiency model** that helped leadership optimize data processing windows, resulting in 25% cloud budget savings and improved decision-making on real-time data usage.
- **Built and globally deployed a diagnostic console** using Django and Kinesis to monitor embedded controller errors. Integrated C++ modules decode CAN streams and transmit diagnostics to the cloud, increasing engineering efficiency by 60% and eliminating on-site machine checks.
- **Architected a customer-facing web platform** to visualize machine paths using silo compaction maps. Extracted and decoded embedded CSV data, transformed it into SQLite3, and rendered NumPy grids as TIFF images—earning 90%+ positive feedback from users.
- **Served as technical lead and strategic architect** across all projects, driving system design, implementation, and cross-functional collaboration with cloud, embedded, and data teams.

Rotation 1

- **Developed a Status Detection module** for display software to monitor and optimize machine performance in real time.
- **Built full stack components** using C++ and Qt, interfacing with embedded systems to capture live signals via CAN (Controller Area Network).
- **Integrated the module into production pipelines**, deploying it as a software artifact for real-time visualization and user interaction.
- **Maintained and extended system infrastructure**, including the signal table, SQL database, CSV logging, and Shell scripts, ensuring reliable data handling and traceability across releases.
- **Enabled real-time automation and diagnostics** by decoding controller signals and integrating them into backend processing and display overlays.
- **Designed and implemented a designer tool** to analyze software deliveries on a quarterly basis and integrate the tool for generating for all production needs, Integrated API and data parsing, deployed globally to software teams using Python. Achieved a 30% increase in positive customer feedback and a 20% rise in usage, for products. Delivering enhancements for Product.

John Deere

Front-end Software Engineer Intern

May 2023 - Aug 2023

- Collaborated with the team to deliver Automation Features in **C++**, while independently developing the Front-end Using **qt frame**, delivered the full functionality, enable user can do the specific action of the product in **user interface**. Calibrating with multiple **algorithm**, communicate with back-end unit **SQL database** to achieve real time unit switch.
- Designed a user interface for automated tasks, enabling easy definition of complex workflows and schedules, enhancing productivity.
- Completed the new feature with a **95%** certification rate, supported by positive co-op feedback.

- Designed an **ios** cutting-edge social platform app for company using **Vue.js**, **React.js**, **HTML** and **CSS**, ensure the functionality of the user interface. Follow the **Figma** requirement of the **UX** design improves quality and quantity of the interface.
- Conducted rigorous testing and quality assurance measures on multiple new applications, proactively addressing technical challenges prior to releasing the beta version on the app store. The app's debut was met with resounding success, attracting nearly 1,000 enthusiastic users on its very first day.

PROJECT

OS development & AI development

Feb 2022 - May 2022

- Developed a custom **Linux kernel** for Intel IA-32 architecture using **C** and **x86 assembly**.
- Implemented filesystem with Paging algorithm to successfully enable the execution of various files and storage.
- Develop a program capable of recognizing digits in images using Convolutional Neural Networks (**CNN**) and **Neural Network** techniques in **Python**.

Software Lead

Sep 2022 - Present

Minerva Game Studio

Madison, WI

- Developed a 2D video game for PC platforms using **Unity**, **C#** with friends, Published in steam.
- Seamlessly integrated in-game **AI**, developed the movement scripts for both NPCs and Boss encounters, enhancing the overall game play experience.
- Successfully rolled out the beta version within the academic environment, garnering acclaim from 90% of players.
- Demonstrated exceptional coding expertise by optimizing the code base, reducing time complexity by more than 50%.
- Demo: https://store.steampowered.com/app/3102950/Library_of_Meialia.

HONORS & AWARDS

Dean's list

UW-Madison, Fall 2021

Academic Excellence Scholarship of Honors School

School of Honor, Every Semester

Heart of Illinois Scholarship

Heart of Illinois, Every Semester